

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal| Lecture 42

Histology of the Nephron & Blood Pressure Control Mechanisms

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Objectives

SOM.MK.1.BPM1.1.CPR.3.HCB.0522	List at least 4 major functions of the kidneys.
SOM.MK.1.BPM1.1.CPR.3.HCB.0523	Briefly recap the gross anatomical structures of a kidney.
SOM.MK.1.BPM1.1.CPR.3.HCB.0524	Differentiate between the renal columns (of Bertin), renal pyramids, cortical labyrinth, renal lobes and lobules based on their components and location.
SOM.MK.1.BPM1.1.CPR.3.HCB.0525	Describe the components of the renal cortex and medulla.
SOM.MK.1.BPM1.1.CPR.3.HCB.0526	Describe the flow of blood through the kidney.
SOM.MK.1.BPM1.1.CPR.3.HCB.0527	Distinguish between a cortical, intermediate and juxtamedullary nephrons based on their structure and location.
SOM.MK.1.BPM1.1.CPR.3.HCB.0528	Identify the components of a urinereforus tubule.
SOM.MK.1.BPM1.1.CPR.3.HCB.0529	List the components of a renal nephron (including the renal corpuscle and tubules).
SOM.MK.1.BPM1.1.CPR.3.HCB.0530	State the function of the renal corpuscle.
SOM.MK.1.BPM1.1.CPR.3.HCB.0531	Describe the microscopic structure of the renal corpuscle.
SOM.MK.1.BPM1.1.CPR.3.HCB.0532	Discuss the structural significance of glomerular capillaries.
SOM.MK.1.BPM1.1.CPR.3.HCB.0533	Discuss the significance of the podocytes.
SOM.MK.1.BPM1.1.CPR.3.HCB.0534	Identify the location of intra-glomerular mesangial cells.
SOM.MK.1.BPM1.1.CPR.3.HCB.0535	Explain the function(s) of intraglomerular mesangial cells.
SOM.MK.1.BPM1.1.CPR.3.HCB.0536	Describe the structures of the glomerular filtration barrier.

Objectives

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Recommended Reading

Histology – A Text and Atlas

Pawlina 8th Edition

Chapter 20: Urinary System Pg. 740-784

[Urinary System | Histology: A Text and Atlas: With Correlated Cell and Molecular Biology, 9e | Premium Basic Sciences | Health Library \(oclc.org\)](#)

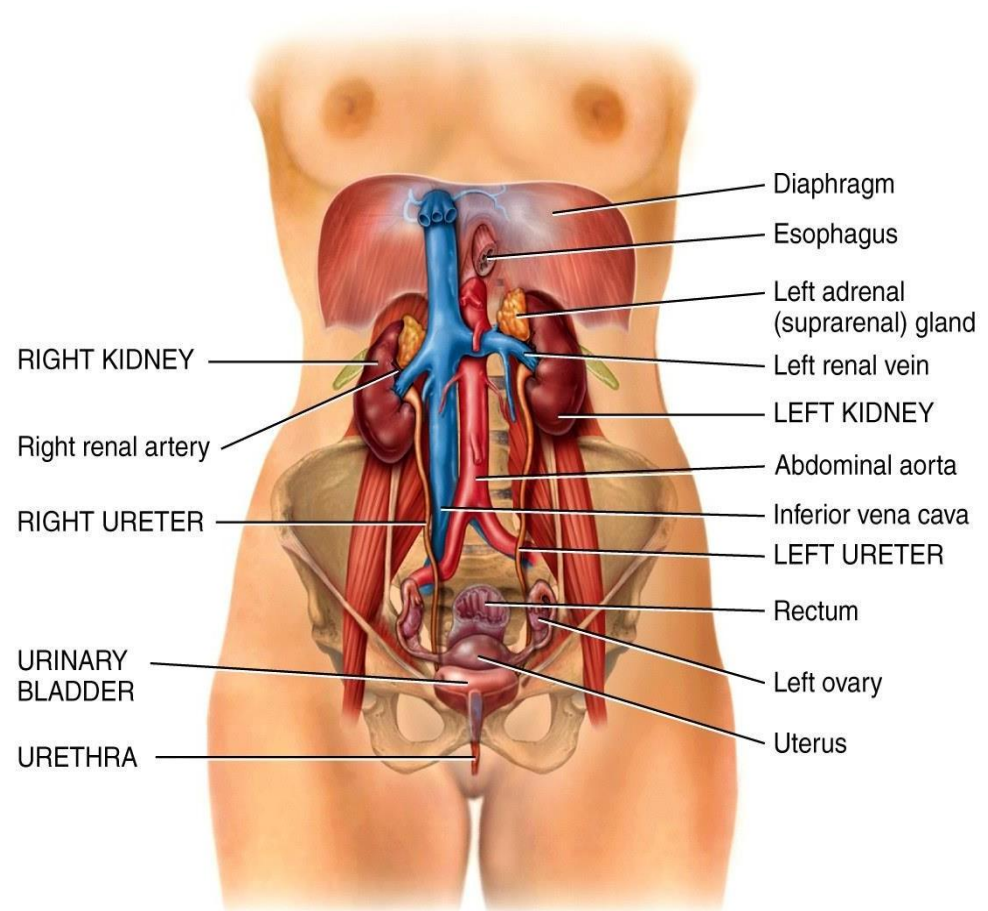


Overview

- **Parts of the Urinary System**
 - **Kidney**
 - **Functions**
 - **Anatomy and Histology**
 - **Renal Lobe**
 - **Renal Lobule**
 - **Blood flow through the Kidney**
 - **Nephron**
 - **Types**
 - **Components:**
 - **Renal Corpuscle**
 - **Proximal Convoluted Tubule**

Urinary System

- Parts:
 - Paired Kidneys
 - Paired Ureters
 - Urinary Bladder
 - Urethra
- Important excretory system
- Formation of urine
- Monitoring of blood pressure



(a) Anterior view

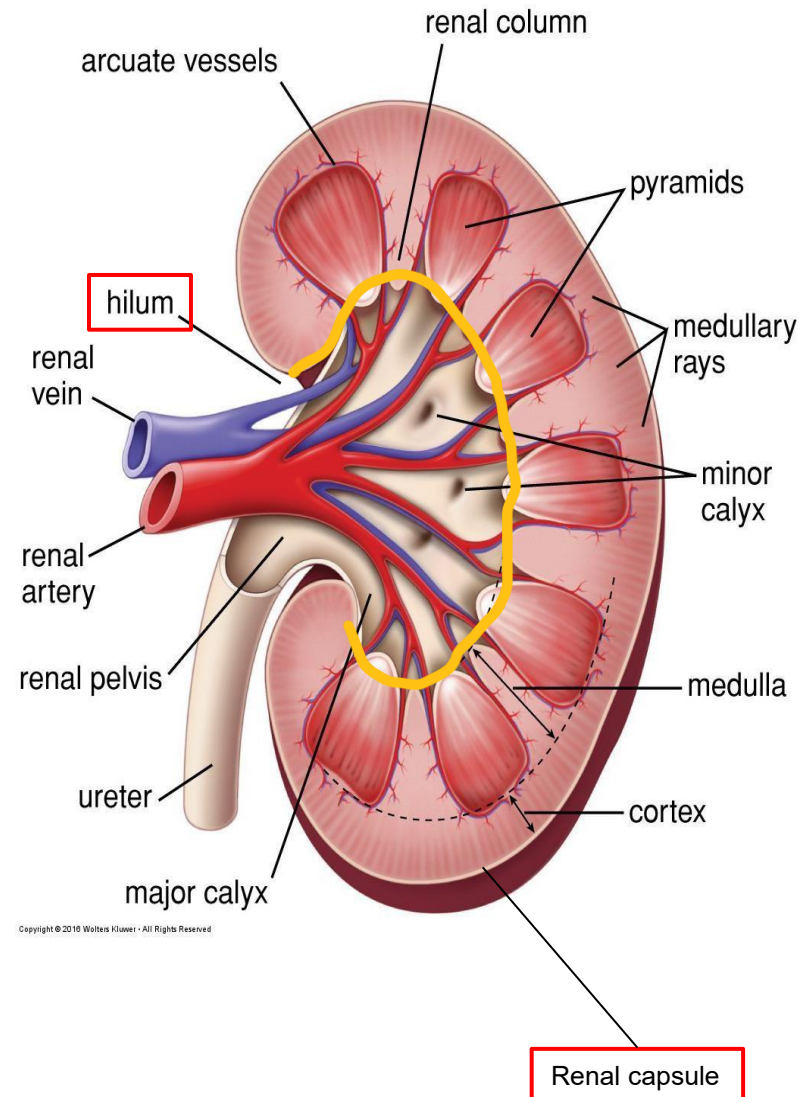
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Kidney – Functions

- Regulation of **acid-base balance**: controls excretion of hydrogen and bicarbonate ions.
- Maintain **composition and volume of extracellular fluid**:
 - Electrolytes: sodium, potassium etc.
 - Water
- **Excretion of**:
 - Metabolic waste: urea, creatinine etc.
 - Bioactive substances (e.g., drugs)
- **Regulation of blood pressure**
- **Endocrine functions**:
 - Hydroxylation of **25-OH vitamin D₃** to hormonally active **1,25- (OH)₂ vitamin D₃**
 - Synthesis and Secretion of **Erythropoietin (EPO)**- acts on bone marrow to regulate erythrocyte formation in response to low blood oxygen concentration.
 - Synthesis and secretion of **renin**- enzyme involved in blood pressure and volume regulation.

Gross Anatomy of the Kidneys

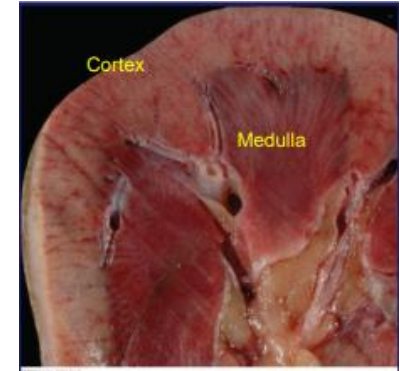
- Paired bean-shaped, retroperitoneal organs on the posterior abdominal wall.
- **Hilum:** located at the concave medial border – the point of entry of neurovascular supply and origin of ureter (renal pelvis).
- **Renal Sinus:** space around the hilar structures, filled with loose connective tissue and adipose tissue.
- **Renal capsule:** connective tissue layer that covers kidney surface:
 - Outer Layer: fibroblasts and collagen fibers.
 - Inner layer: myofibroblasts



Gross Anatomy of the Kidneys

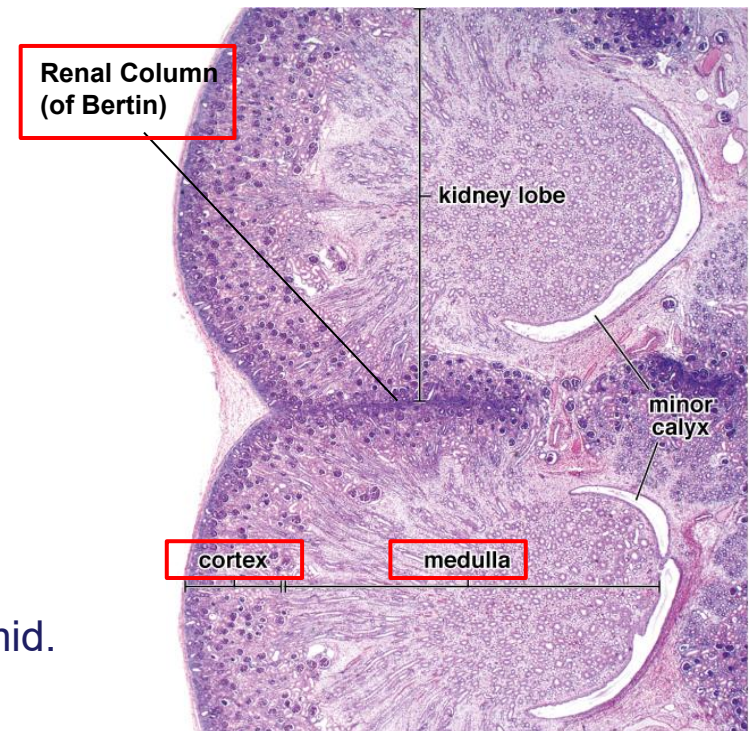
Renal Cortex:

- Lies immediately beneath the renal capsule.
- Granular appearance: convoluted components of the nephron.



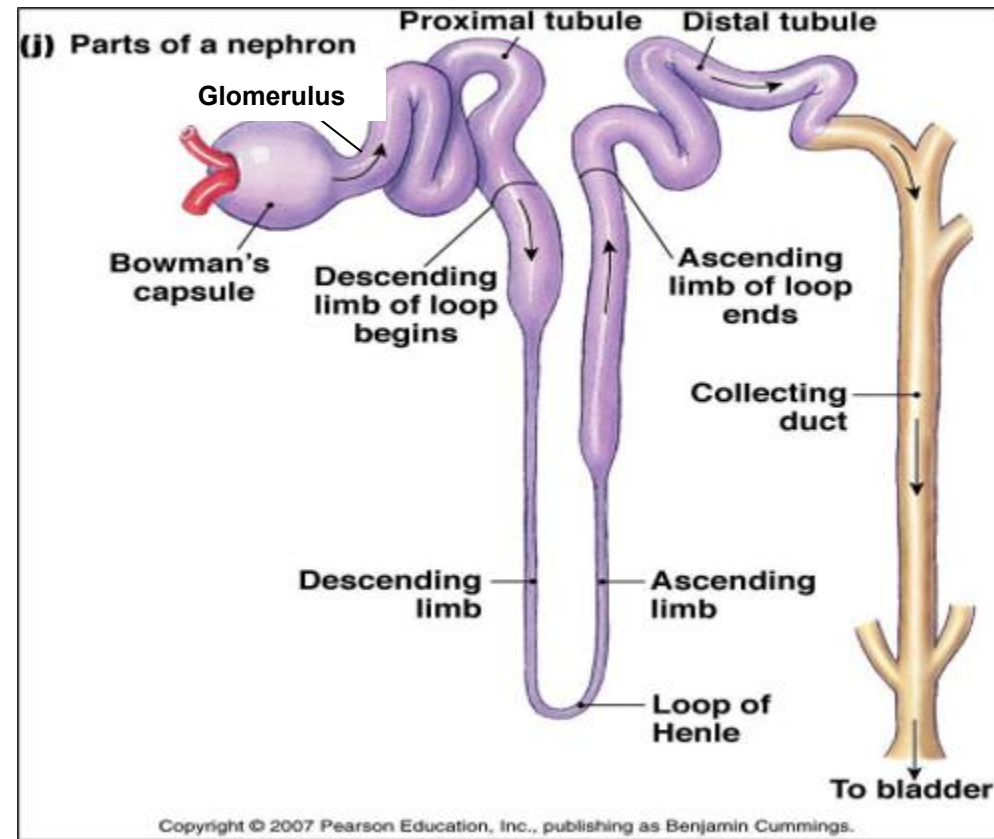
Renal Medulla:

- 8 – 12 **renal pyramids**.
- Apex of pyramid (**papilla**) projects into minor calyx.
- Striated appearance: straight portions of the tubules and straight blood vessels.
- Divided into **outer and inner medulla**.
- **Renal Columns(of Bertin)**: cortical tissue extend around the lateral portion of the pyramid.
Considered part of the medulla.



The Renal Nephron

- The functional unit of the kidney.
- **Renal Corpuscle:**
 - **Glomerulus**
 - **Bowman's Capsule**
- **Renal Tubules**
 - **Proximal Convoluted**
 - **Loop of Henle (straight portions):**
 - **Thick Descending/Proximal Straight.**
 - **Thin descending**
 - **Thin Ascending**
 - **Thick Ascending/Distal Straight**
 - **Distal Convoluted**
- **Uriniferous tubule: Nephrons + connecting tubule.** Connecting tubule connects the nephron to the collecting duct.



Types of Nephron

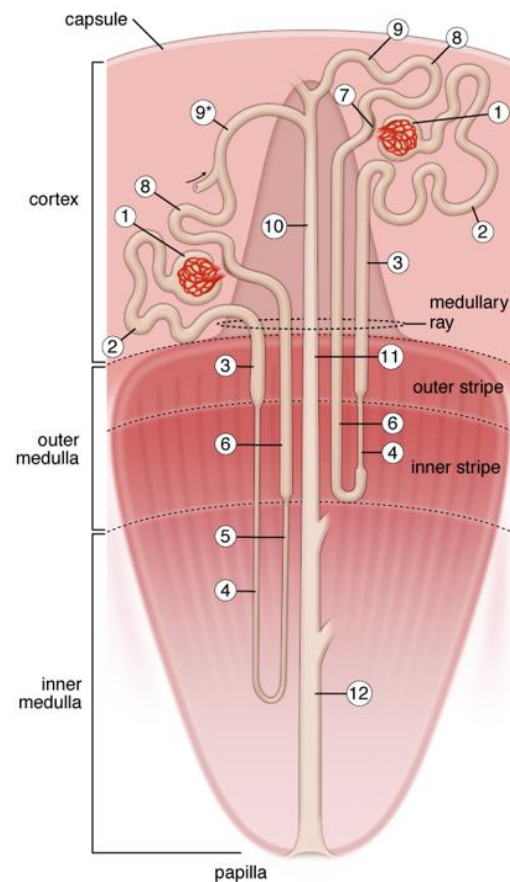
• Cortical

- Short loops of Henle extending only into the outer medulla (hairpin turn created by the distal straight tubule)

• Intermediate (mid cortical)

• Juxtamedullary

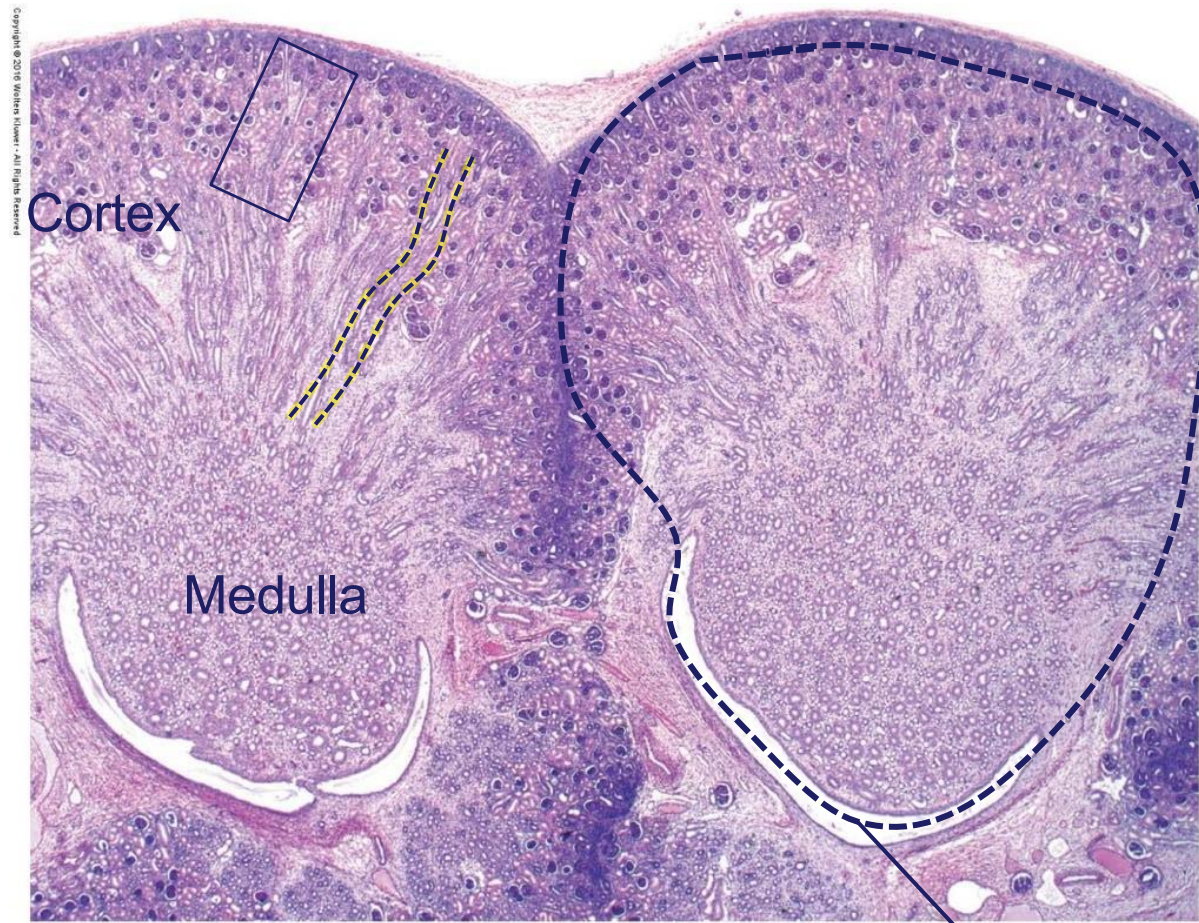
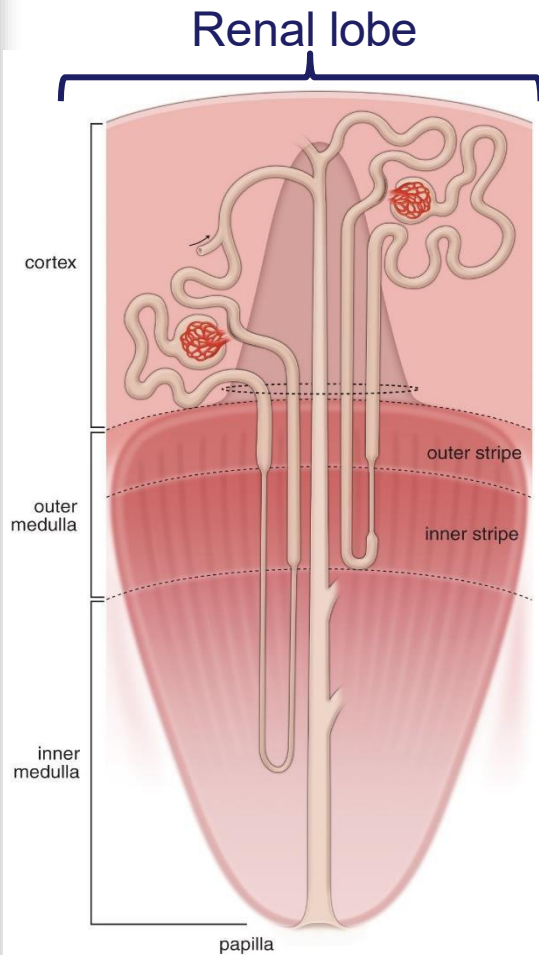
- Renal corpuscles occur close to the base of the pyramid.
- Long loops of Henle and long ascending thin segments that extend into the inner medulla.
- These are crucial for concentrating urine.



- 1: Renal Corpuscle
- 2: Proximal Convoluted Tubule
- 3: Proximal Straight Tubule/Thick Descending
- 4: Thin descending limb of the loop of Henle
- 5: Thin ascending limb of the loop of Henle
- 6: Distal Straight Tubule/Thick Ascending
- 7: Juxtaglomerular Apparatus – Macula Densa
- 8: Distal Straight Tubule
- 9: Connecting Tubule
- 10: Cortical Collecting Duct
- 11: Outer Medullary Collecting Duct
- 12: Inner Medullary Collecting Duct

Renal Lobe

Renal Lobe: consists of the medulla and its associated cortex.



Renal lobe

Renal Lobule

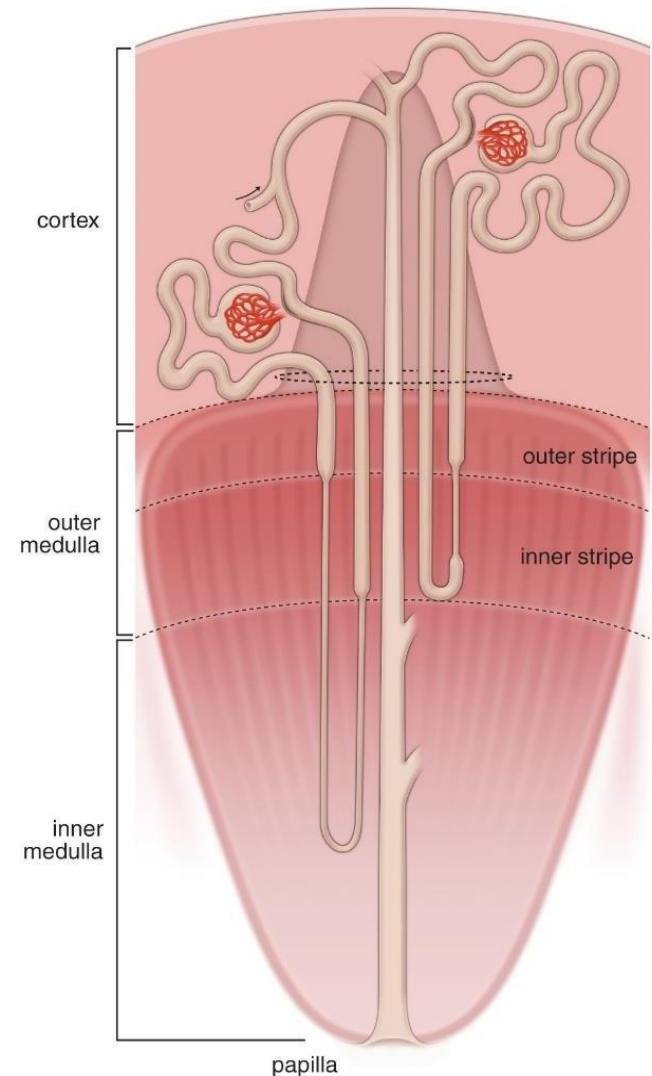
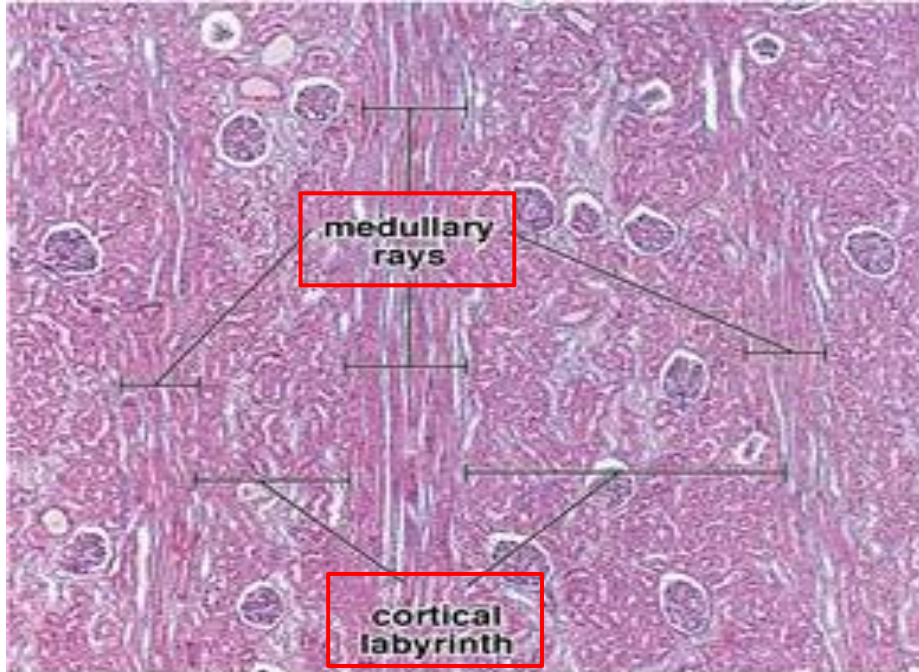
- Consists of a collecting duct and the nephrons it drains.

1. Medullary ray (Pars recta):

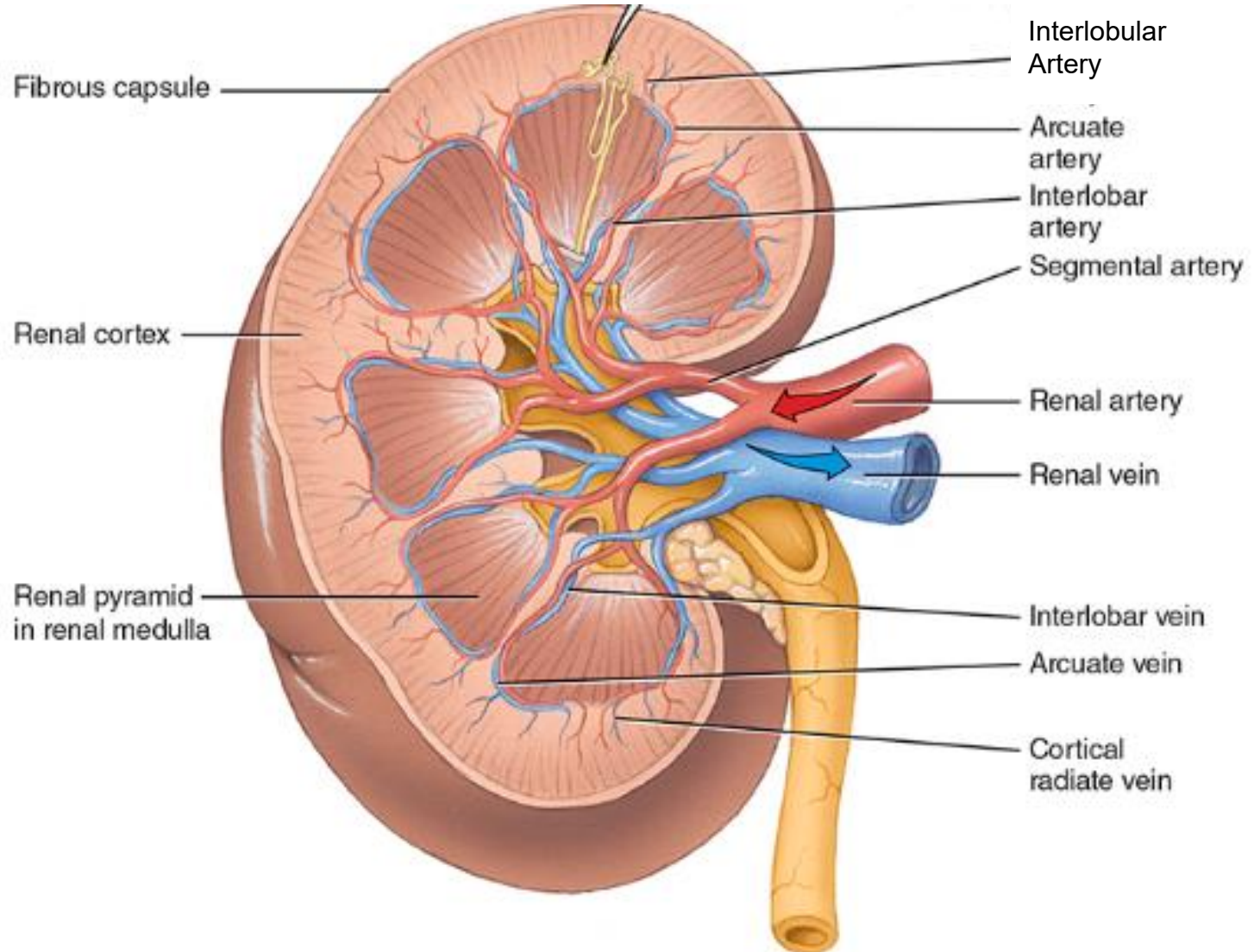
- Straight tubules and collecting ducts.

2. Cortical Labyrinth (Pars convoluta):

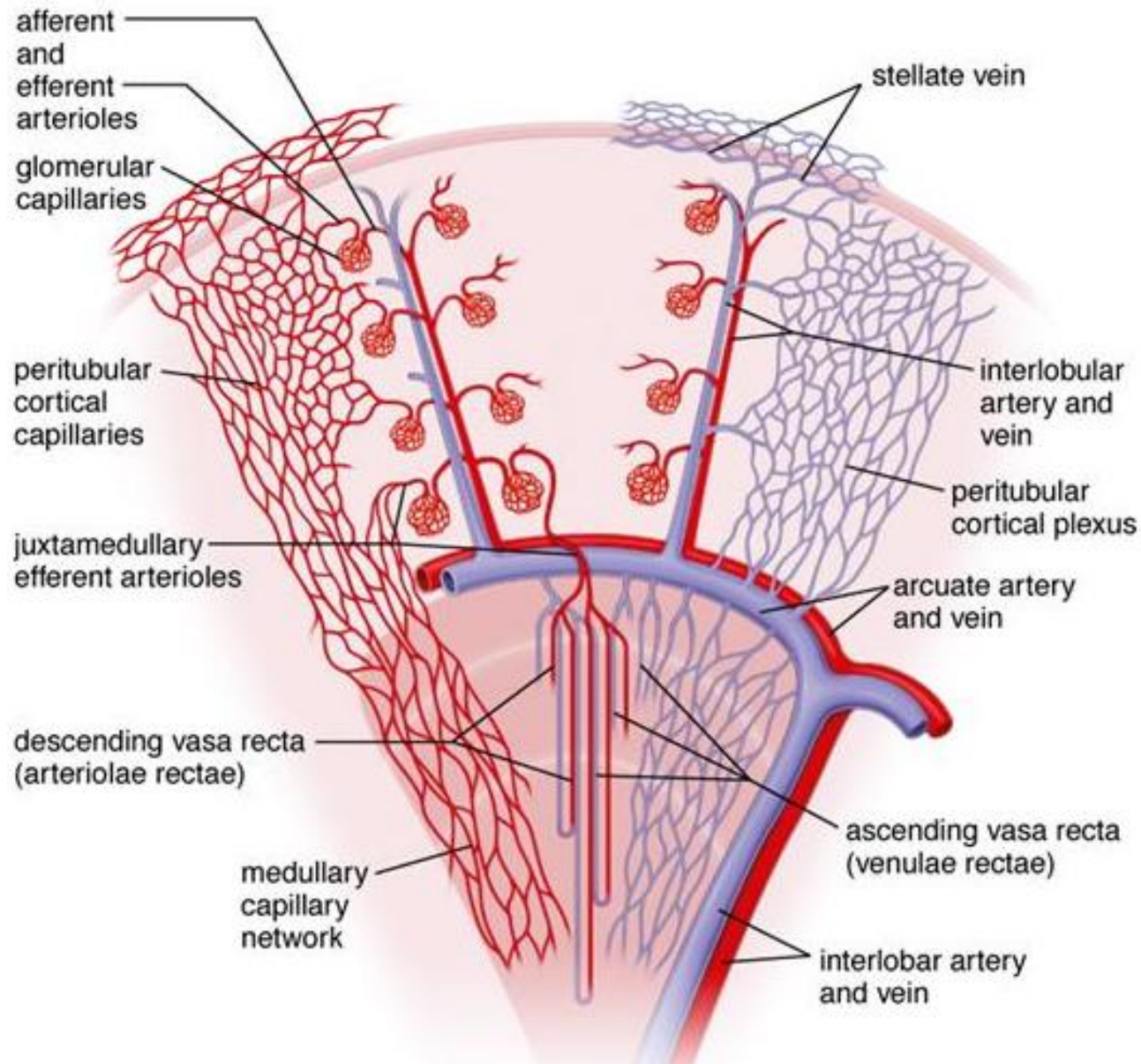
- Renal corpuscle and proximal and distal convoluted tubules (proximal and distal)



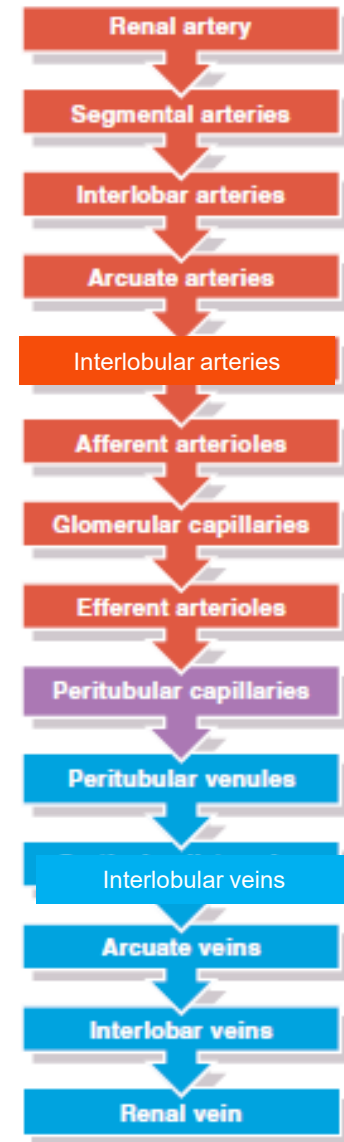
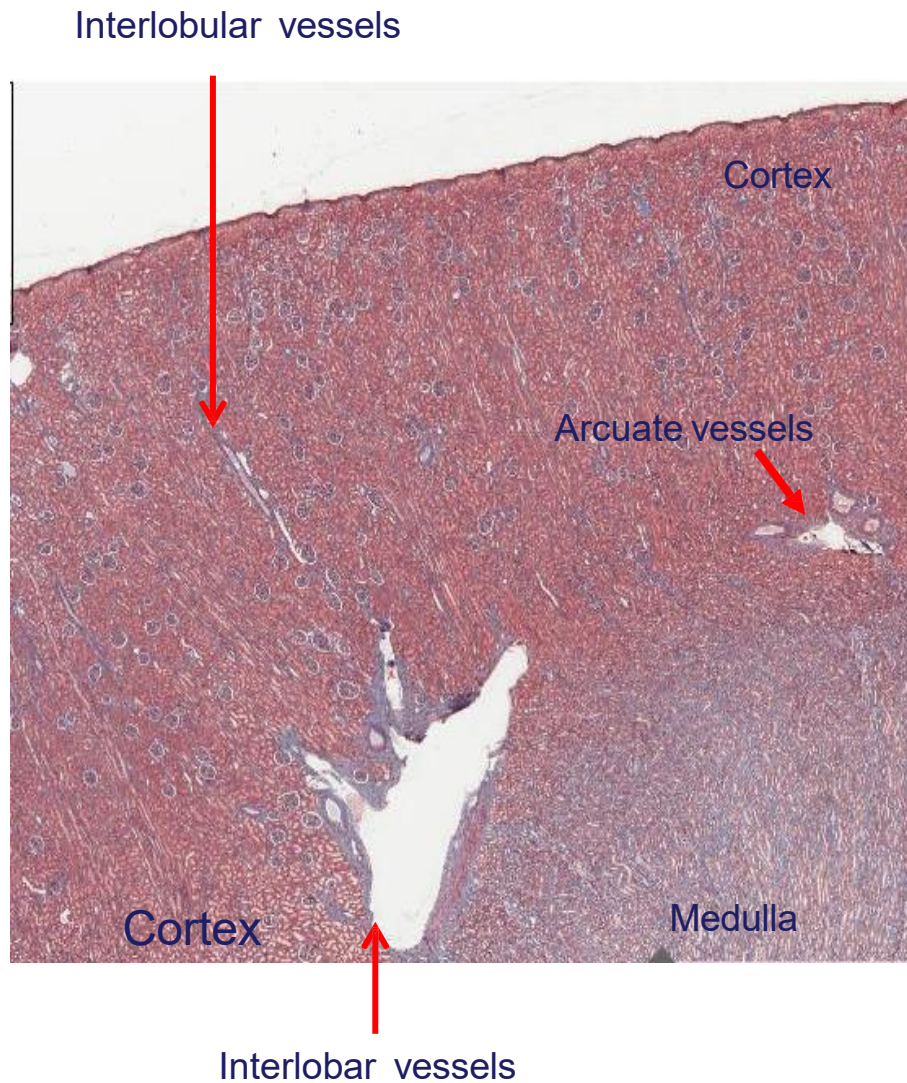
Blood Flow through the Kidney



Blood Flow through the Kidney



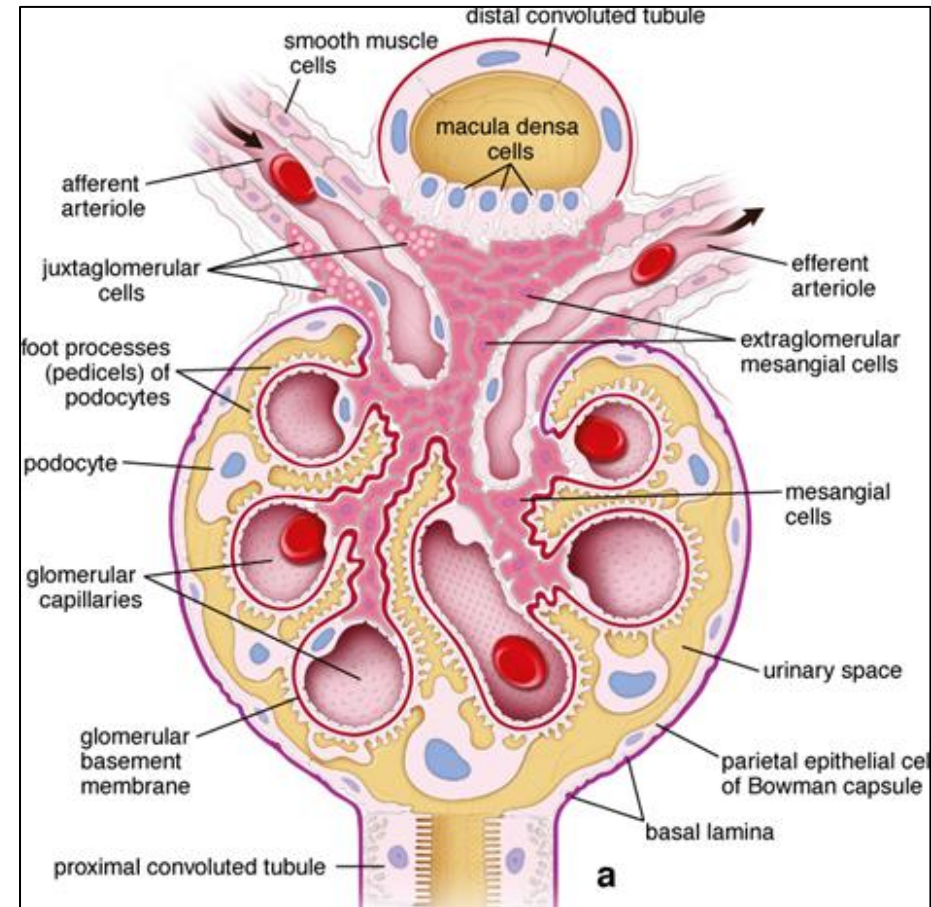
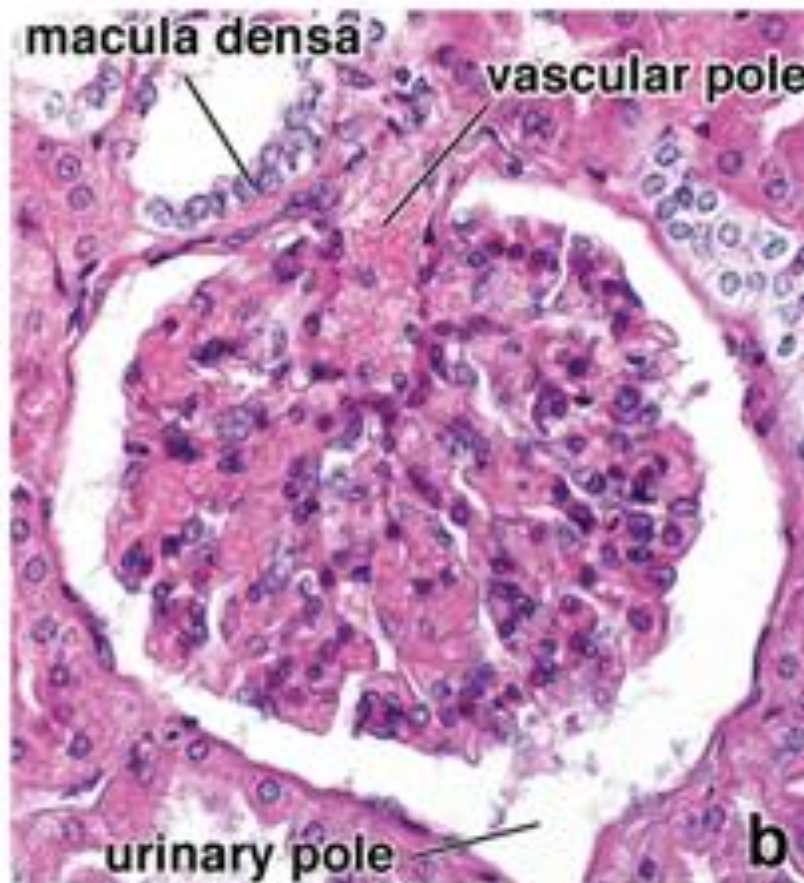
Blood Flow through the Kidney



The Nephron: Renal Corpuscle

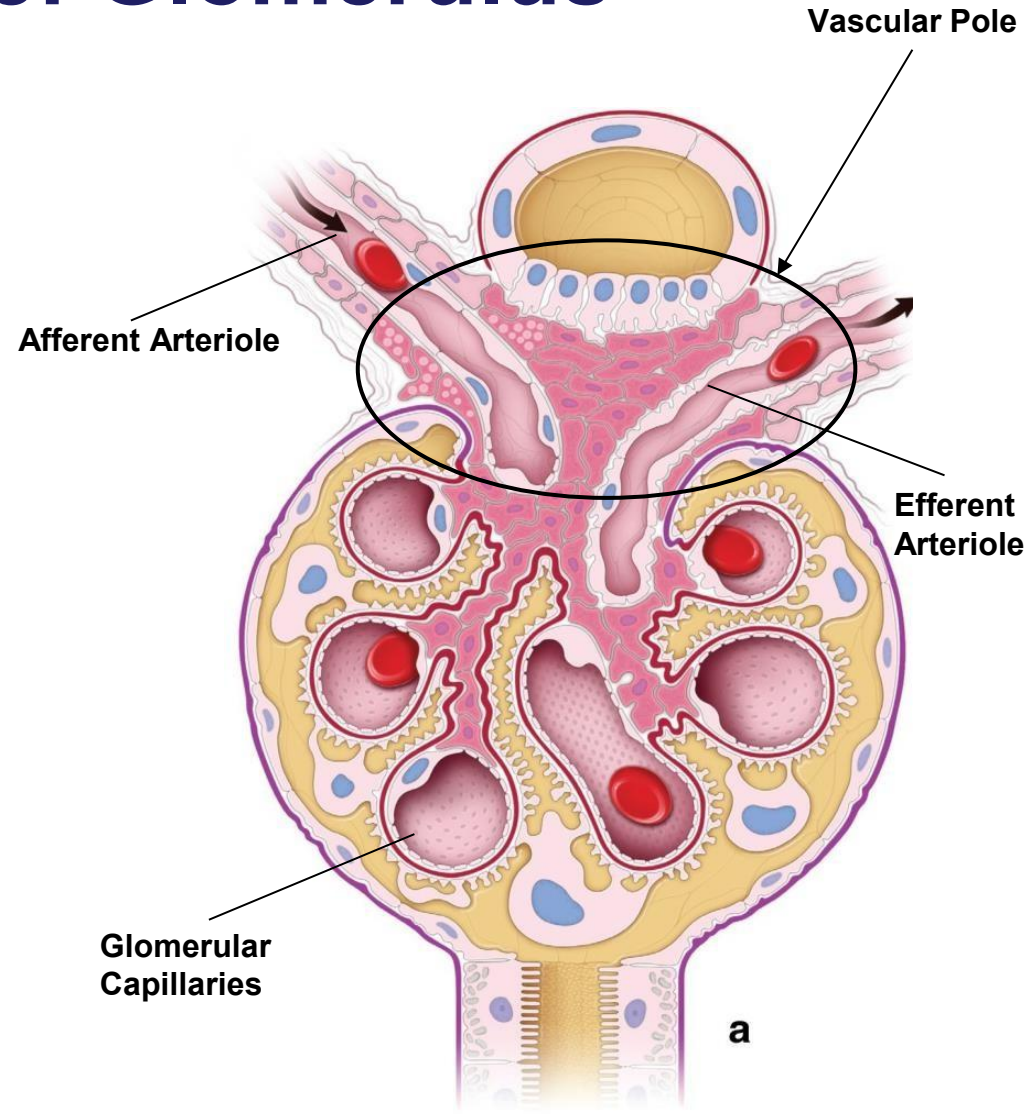
Functions: responsible for the filtration function of the kidneys.

Components: 1. **Glomerulus**, 2. **Bowman's capsule**, 3. **Mesangium**



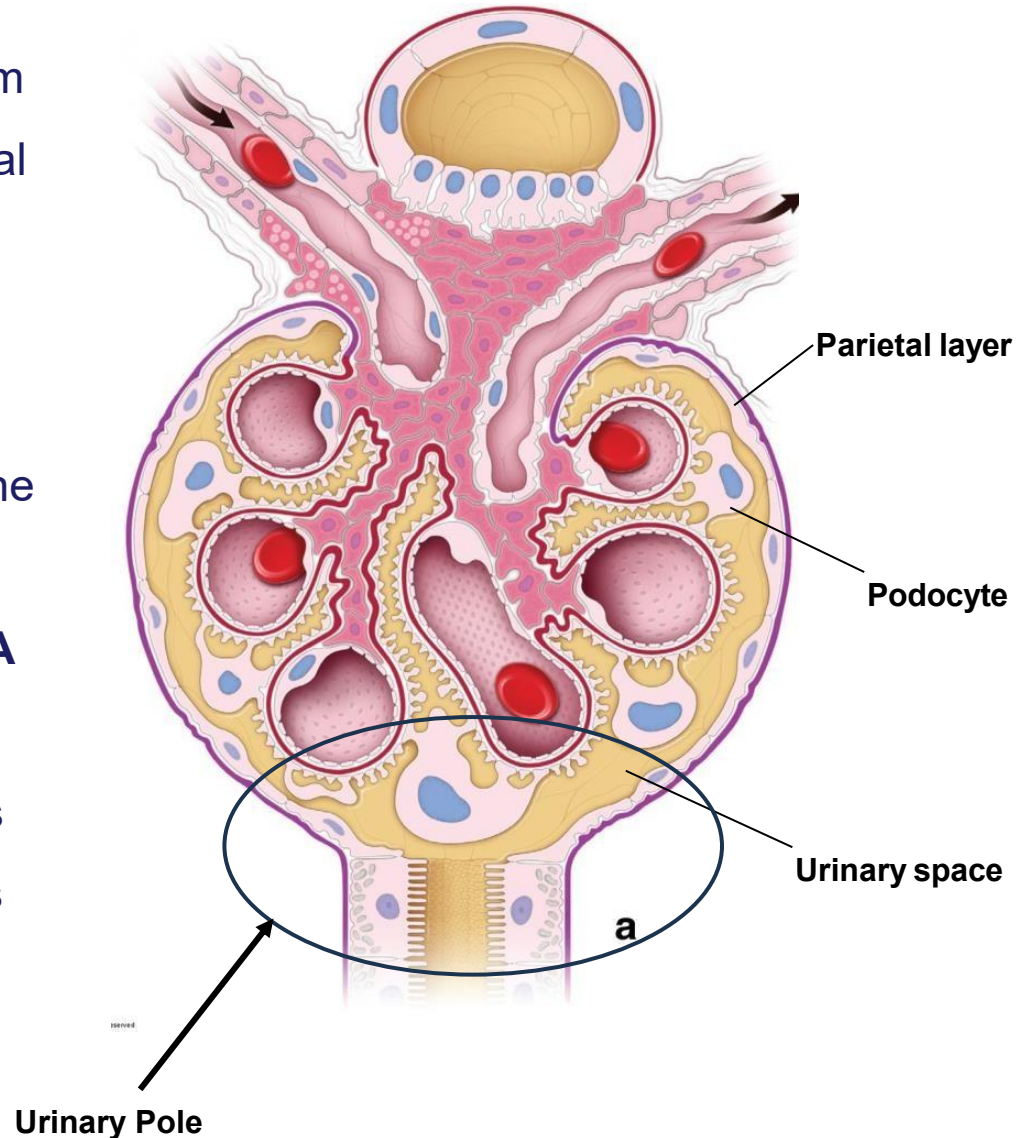
Renal Corpuscle: Glomerulus

- Tuft of **fenestrated capillaries** composed of 10 – 20 capillary loops.
- Supplied by an **afferent arteriole**.
- Drained by an **efferent arteriole**.
- **Vascular Pole**: site where the afferent and efferent arterioles penetrate and exit the Bowman's capsule.



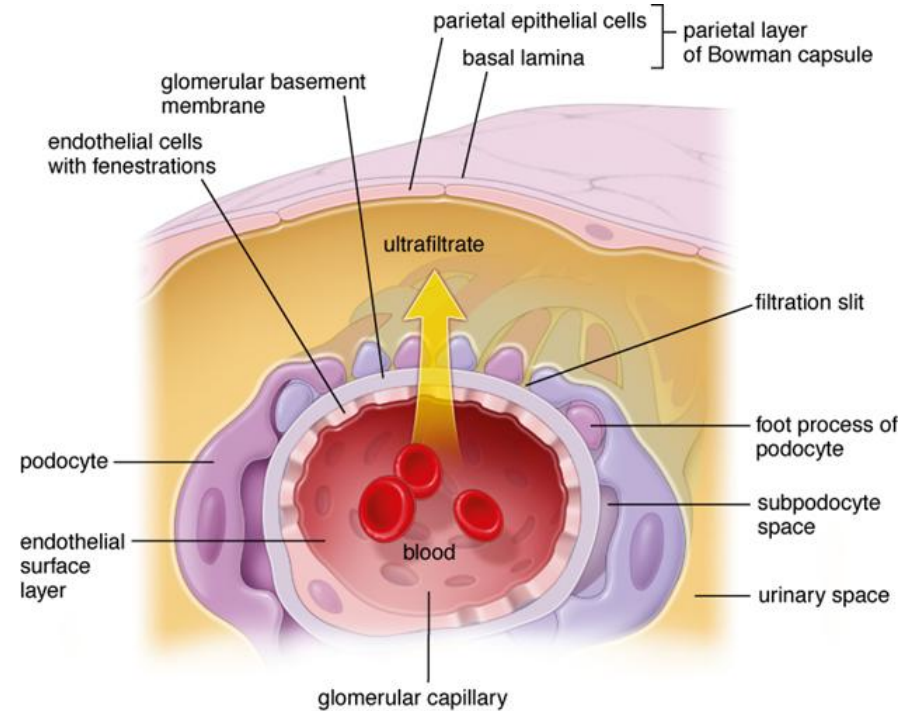
Renal Corpuscle: Bowman's Capsule

- **Parietal Layer – Outer layer**
 - Simple squamous epithelium
 - Continuous with the proximal convoluted tubule.
- **Visceral layer – Podocytes**
 - Modified squamous cells
 - Processes extend around the glomerular capillaries.
- **Urinary Space (Bowman's AKA Capsular Space)**
 - Lies between the two layers
 - Collects ultrafiltrate which is directed to the PCT at the **urinary pole**.



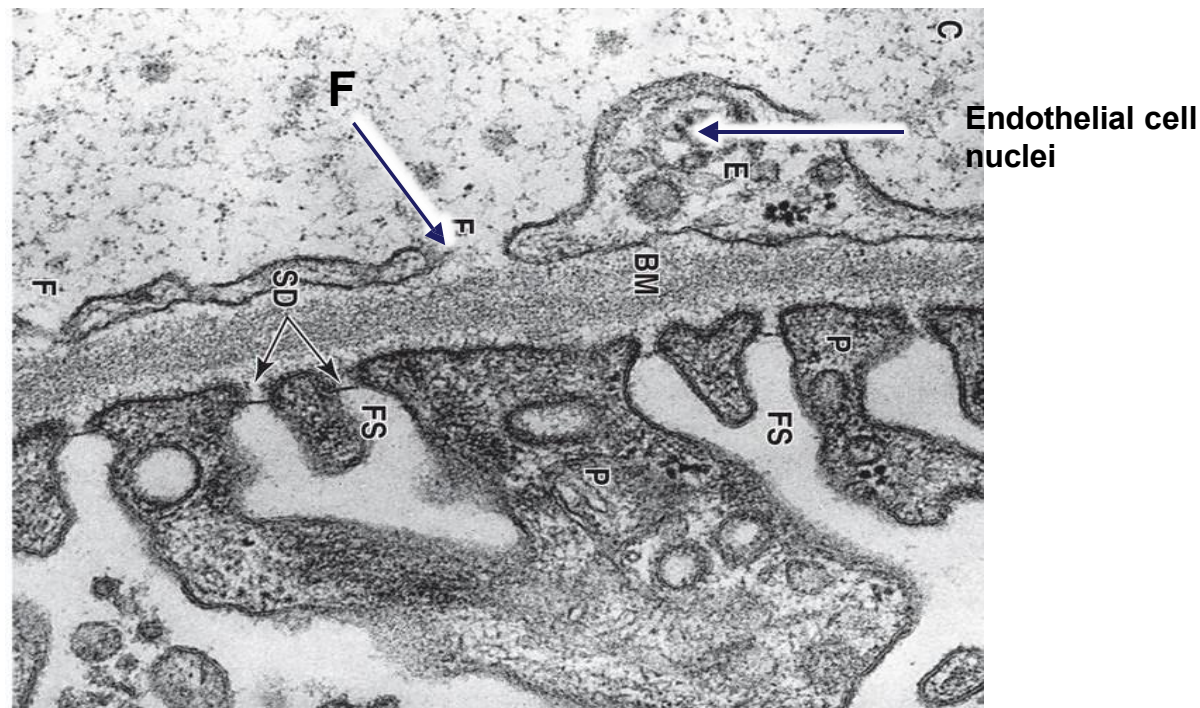
The Filtration Barrier

- Enclosed by the parietal layer of the Bowman's capsule.
- Consists of 3 components:
 1. **Endothelium of the glomerular capillaries**
 2. **Glomerular Basement Membrane**
 3. **Visceral layer of the Bowman's Capsule.**



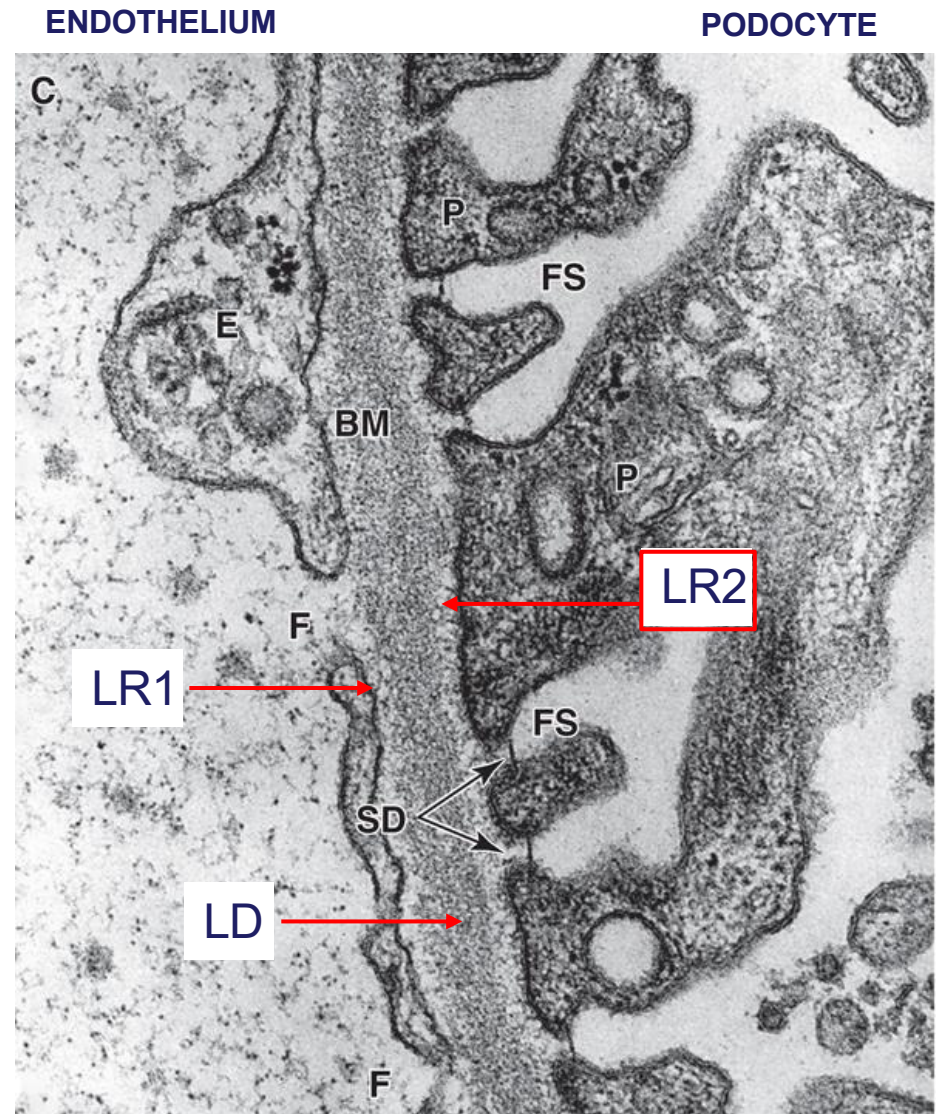
The Filtration Barrier: Endothelium of Glomerular Capillaries

- Numerous fenestrations (F) >70-90 nm
- **No diaphragms**
- Produces nitric oxide and prostaglandins (PGE_2).
- Abundant aquaporin-1 receptors allows for fast movement of water through the epithelium.



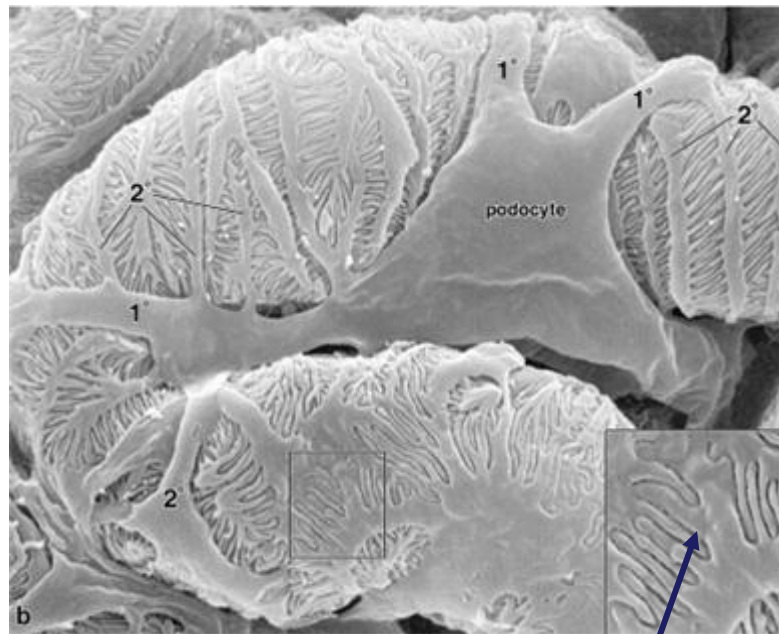
The Filtration Barrier: Basement Membrane

- Thick 300-370 nm
- Fused basal lamina of endothelial cells and podocytes.
- Molecular structure:
 - **Type IV collagen**
 - **Heparan Sulphate**
- Structural arrangement:
 - **Lamina rare interna (LR1):** of the capillary
 - **Lamina densa (LD):** rich in Type IV collagen
 - **Lamina rare externa (LR2):** of the podocyte, rich in heparan sulfate.

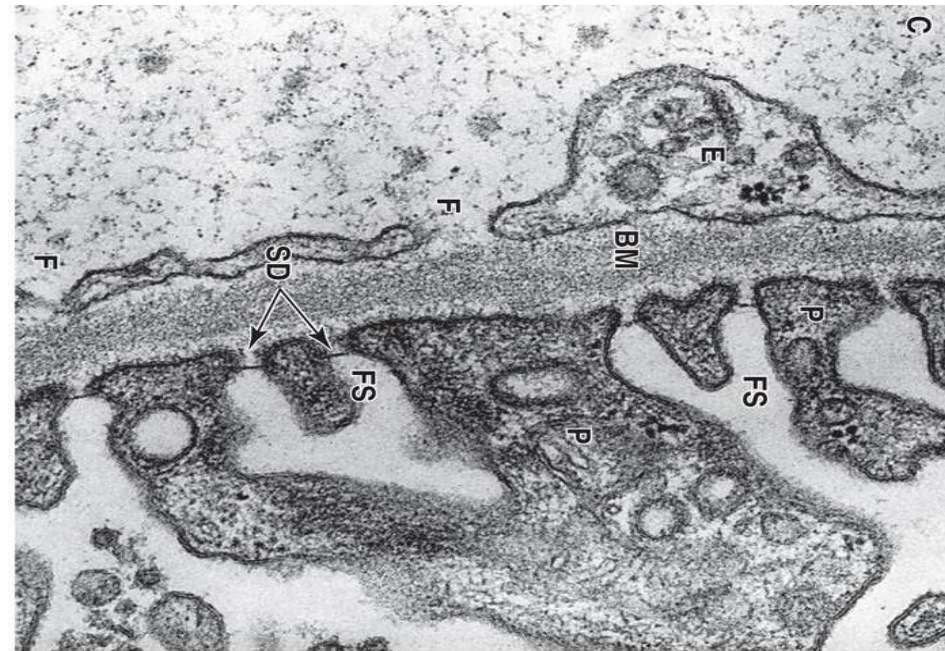


The Filtration Barrier: Visceral Layer of Bowmans Capsule

- Modified epithelial cells called podocytes, which are highly ramified.
 - Primary processes > secondary processes > pedicles/foot processes.
- Filtration slits (FS) found between the pedicels: 40nm
 - Covered by slit diaphragm (SD).

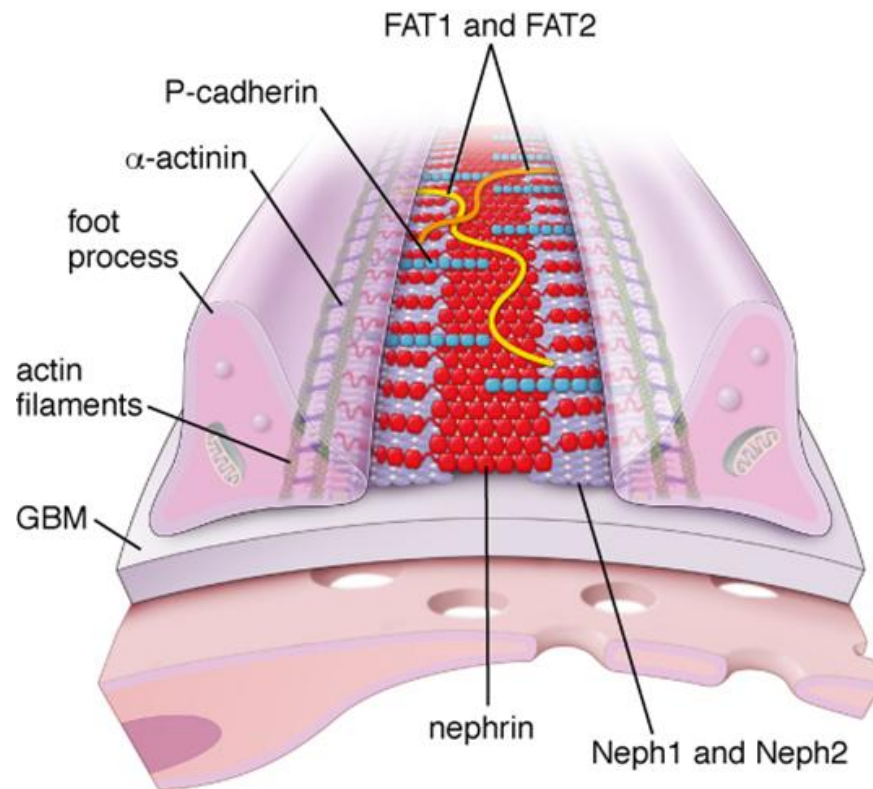


Filtration slits; alternating pedicles belonging to the secondary process of two different cells.



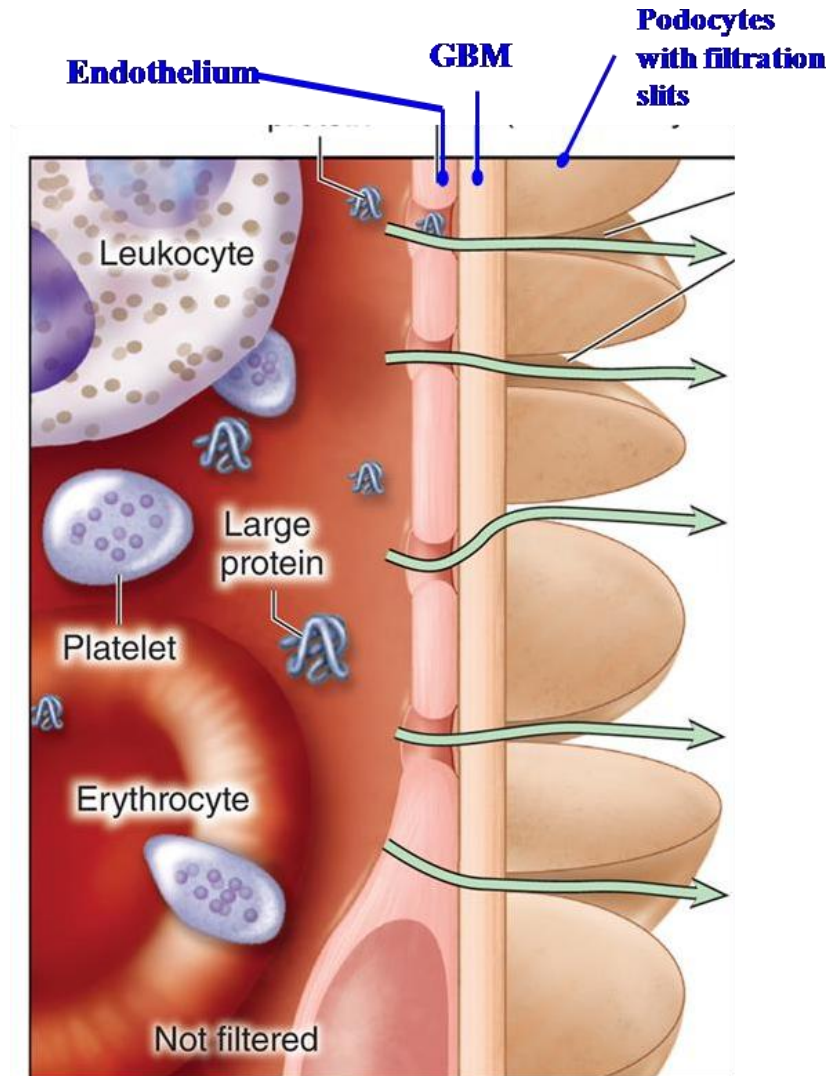
Visceral Layer of Bowmans Capsule: Filtration Slit

- **Zipper-like thin sheet** with a dense center containing transmembrane protein **nephrin** (from opposite foot processes).
- Anchored to **actin filaments** within the pedicles of podocytes.
- Other adhesion proteins include **Neph-1**, **Neph-2**, **P-cadherin**, **FAT1** and **FAT-2**.



Factors that affect Filtration

1. **Charge:** Negatively charged molecules are repelled.
 - Achieved by the heparan sulfate in lamina rare externa.
2. **Size:** $\geq 70,000$ Daltons (70 kilodaltons) molecular weight cannot pass through e.g., albumin.
 - Achieved by both the basement membrane and the filtration slit diaphragm.
3. **Shape:** >3.6 nm radius e.g., blood cells cannot pass through.
 - Achieved by the basement membrane.



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Clinical Correlates

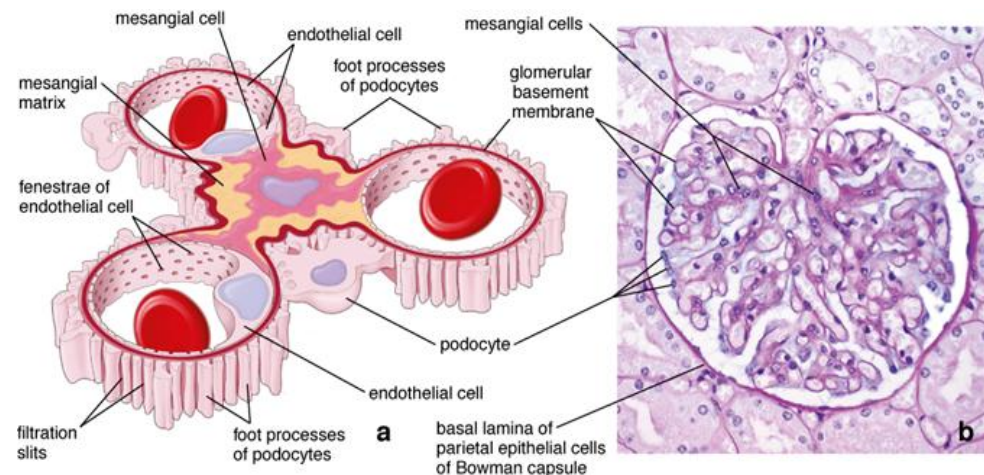
Disease	Pathology	Features	Histological Changes
Anti-GBM antibody-induced Glomerulonephritis	Autoimmune response to type IV collagen in the glomerular basement membrane (GBM) creating anti-GBM antibodies. Triggered by viruses, cancer, pharmacologic agents and chemical compounds.	• Hematuria • Proteinuria • Progressive renal failure	Crescent-shaped glomerulus
Goodpasture Syndrome	In some individuals, anti-GBM antibodies may also cross-react with the alveolar basement in the lungs.	• Hematuria • Proteinuria • Progressive renal failure. • Pulmonary Symptoms: Shortness of breath, cough, bloody sputum	Crescent-shaped glomerulus
Alport Syndrome	Mutation in the gene encoding the alpha 5 chain of type IV collagen. It is hereditary.	• Hematuria • Proteinuria • Progressive renal failure	Glomerular basement membrane becomes irregularly thickened with laminated lamina densa and fails to serve as an effective filtration barrier.

Renal Corpuscle: Mesangium

1. Mesangial Cells:

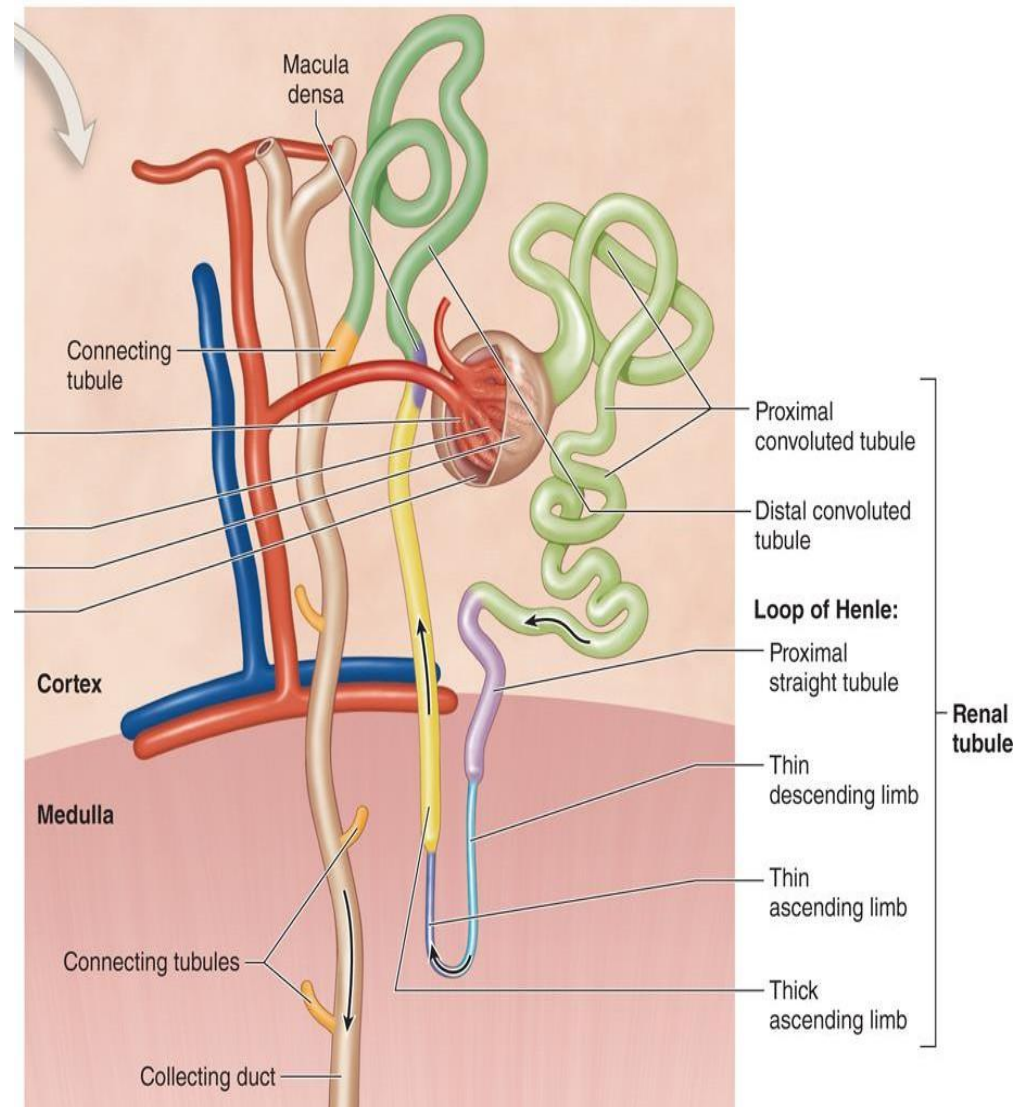
- Extraglomerular/Lacis
- **Intraglomerular**
 - **Phagocytosis** of residue along the GBM and filtration slit diaphragm to keep the glomerular filter free of debris.
 - **Structural support:** secretes extracellular mesangial matrix.
 - **Secretion of inflammatory substances:**
 - Interleukin-1, prostaglandin E₂, PDGF.
 - **Modulation of glomerular distension:**
 - Contractile properties may regulate glomerular distension in increased blood pressure.
 - Response to angiotensin II & atrial natriuretic factor/ANF.

2. Matrix: provide support for the Podocytes in areas where the epithelial basement membrane is absent or incomplete.



Nephron: Tubules

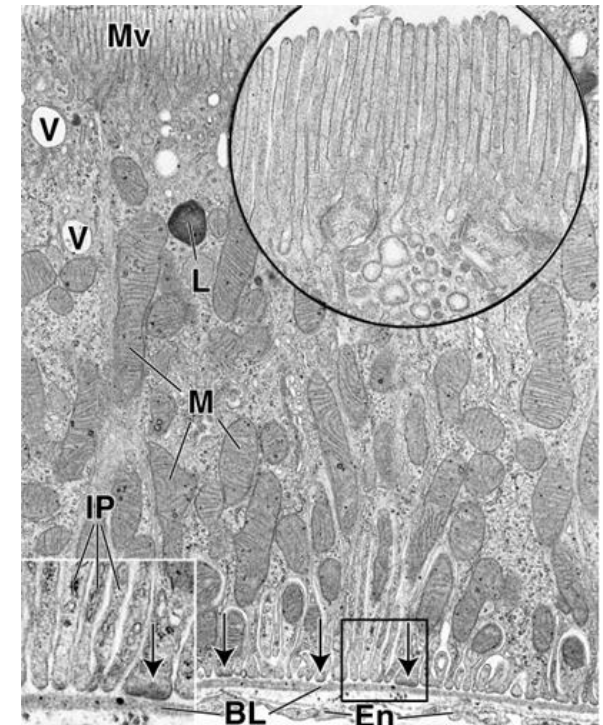
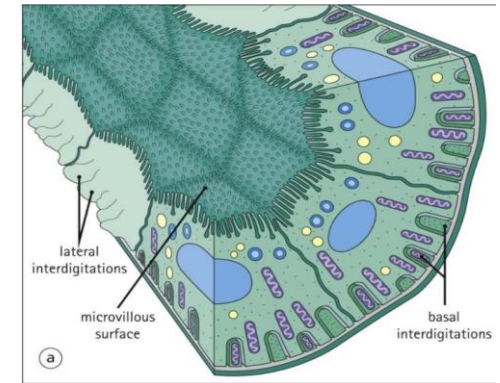
- Renal tubules modify glomerular ultrafiltrate.
 - Reabsorption
 - Partially
 - Completely
 - Secretion
- These processes reduces the volume of the ultrafiltrate, and the urine is made hyperosmotic.



Proximal Convoluted Tubules

Key Identifying histological features:

- **Predominant tubular profile in cortex.**
- **Simple cuboidal epithelium:**
 - **Large cells:** large spaces between adjacent cellular nuclei.
 - **Apical Surface:** extensive microvilli which forms a brush border; gives a dirty lumen appearance.
 - **Basolateral Surface:**
 - **Tight junctions and zonula adherens**
 - **Plicae:** folds on the lateral surface which interdigitate with adjacent cellular processes.
 - **Basal interdigitations**
 - **Basal striations** associated with elongated mitochondria; makes the PCT more eosinophilic than DCT.

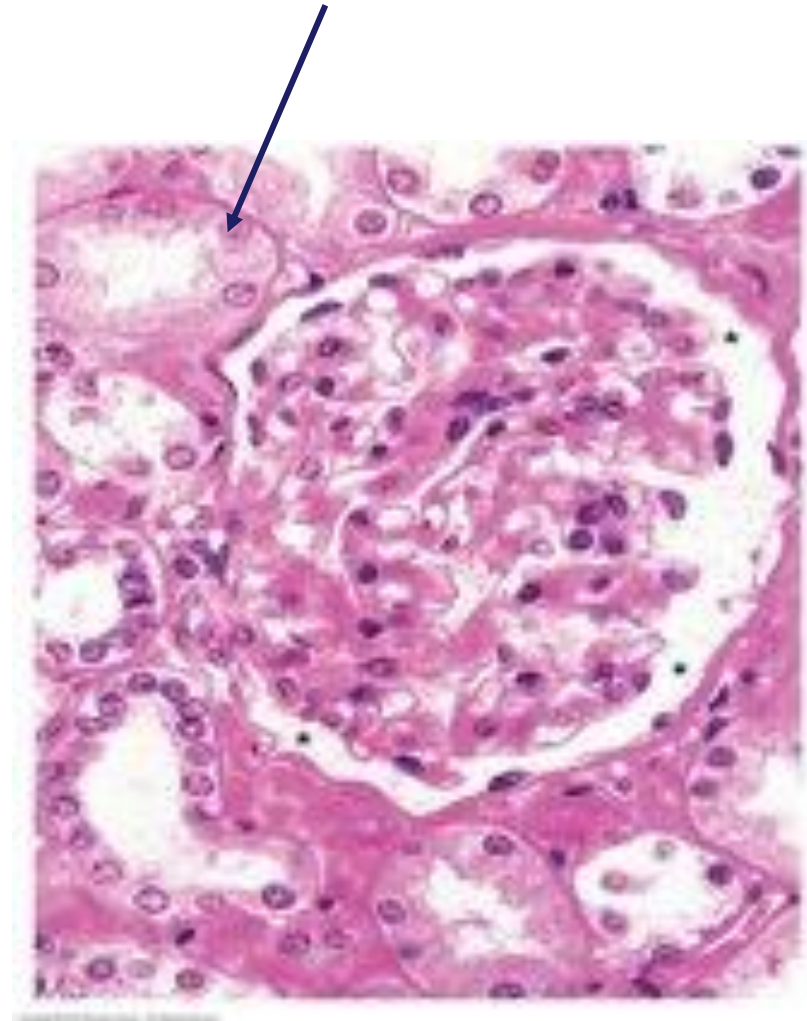


Proximal Convoluted Tubules

Functions:

- Receive the **ultra-filtrate from the urinary space** and **reabsorbs 65%** of it.
 - **100% of glucose**
 - **98% amino acids, small polypeptides.**
 - Water:
 - Na^+/K^+ -ATPase pumps: on the lateral folds create osmotic gradient for water reabsorption.
 - Aquaporin-1 (AQP-1): small transmembrane protein that functions as molecular water channel.
 - **Some proteins and large peptides are returned to the blood via endocytosis.**

Proximal Convoluted Tubule





Overview

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